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Sub ject : DSA

Unit 2, S11, SLO1 - Doubly Linked List – Deletion at Beginning and End

Algorithm to delete the first node

Step 1: IF START = NULL

Write UNDERFLOW

Go to Step 6

[END OF IF]

Step 2: SET PTR = _____

Step 3: SET START = _____

Step 4: SET _____ = NULL

Step 5: FREE _____

Step 6: EXIT

Answer: Step 2: SET PTR = START

Step 3: SET START = START -> NEXT

Step 4: SET START -> PREV = NULL

Step 5: FREE PTR

Step 6: EXIT

Algorithm to delete the last node

Step 1: IF START = NULL

Write UNDERFLOW

Go to Step 7

[END OF IF]

Step 2: SET PTR = _____

Step 3: Repeat Step 4 while _____ != NULL

Step 4: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 5: SET _____ = NULL

Step 6: FREE _____

Step 7: EXIT

Answer: Step 2: SET PTR = START

Step 3: while PTR -> NEXT != NULL

Step 4: SET PTR = PTR -> NEXT

Step 5: SET PTR -> PREV -> NEXT = NULL

Step 6: FREE PTR

Step 7: EXIT